Fundamentals of Numerical Analysis

Numerical methods

1) Bisection method:

$x_n = \frac{a+b}{2}$

Q. Find the root of $x^3 - 4x - 9 = 0$

(i) $f(x) = x^3 - 4x - 9$

(ii) Find an interval by trial,

$f(0) = 0^3 - 4 \times 0 - 9 = -9$

$f(1) = 1^3 - 4 \times 1 - 9 = -12$

$f(2) = 2^3 - 4 \times 2 - 9 = -9$ (-ve)

$f(3) = 3^3 - 4 \times 3 - 9 = 6$ (+ve)

Interval $[a, b] = [2, 3]$

(iii) let $x_1 = \frac{a+b}{2} = \frac{2+3}{2} = 2.5$

$\Rightarrow f(x_1) = (2.5)^3 - 4 \times (2.5) - 9 = -3.375$ (-ve)

note: 2 is replaced by 2.5 because they are the -ve values.

changing interval $[a, b] = [2.5, 3]$

$x_2 = \frac{2.5+3}{2} = 2.75$

$\Rightarrow f(x_2) = 0.796$

$[a, b] = [2.5, 2.75]$ $\Rightarrow x_3 = \frac{2.5 + 2.75}{2} = 2.625$

$\Rightarrow f(x_3) = 1.648^3 - 4 \times 1.648 - 9 = -1.41$

$[a, b] = [2.625, 2.75]$ $\Rightarrow x_4 = 2.687$

$\Rightarrow f(x_4) = -0.33$

$[a, b] = [2.687, 2.75]$ $\Rightarrow x_5 = 2.718$ $\Rightarrow f(x_5) = 0.207$

$[a, b] = [2.687, 2.718]$ $\Rightarrow x_6 = 2.70$ $\Rightarrow f(x_6) = -0.117$

$[a, b] = [2.70, 2.718]$ $\Rightarrow x_7 = 2.709$ $\Rightarrow f(x_7) = 0.044$

$[a,b] = [2.70, 2.709] \Rightarrow x_8 = 2.7045 \Rightarrow f(x_8) = -0.03642$

$[a,b] = [2.7045, 2.709] \Rightarrow x_9 = 2.7067 \Rightarrow f(x_9) = 4.19 \times 10^{-3}$

$[a,b] = [2.7045, 2.7067] \Rightarrow x_{10} = 2.7056 \Rightarrow f(x_{10}) = -0.016$

$[a,b] = [2.7056, 2.7067] \Rightarrow x_{11} = 2.70615 \Rightarrow f(x_{11}) = -6.7 \times 10^{-3}$

Q. Solve $x^3 - x - 2 = 0$ correct to 3 iterations.

$f(x) = x^3 - x - 2 \Rightarrow f(0) = -2$

$f(1) = -2$

$f(2) = 4$

~~$[a,b] = [2,4] \Rightarrow x_1 = 3 \Rightarrow f(x_1) = 22$~~

~~$[a,b] = [2,22] \Rightarrow x_2 =$~~

$[a,b] = [1,2] \Rightarrow x_1 = 1.5 \Rightarrow f(x_1) = -0.125$

$[a,b] = [1.5,2] \Rightarrow x_2 = 1.75 \Rightarrow f(x_2) = 1.609$

$[a,b] = [1.5, 1.75] \Rightarrow x_3 = 1.625 \Rightarrow f(x_3) = 0.666$

$[a,b] = [1.5, 1.625] \Rightarrow x_4 = 1.562 \Rightarrow f(x_4) = 0.249$

$[a,b] = [1.5, 1.562] \Rightarrow x_5 = 1.531 \Rightarrow f(x_5) = 0.057$

$[a,b] = [1.5,$

HW.

1Q. Solve $x^3 - 2x - 5 = 0$ using bisection method (4 iteration) [ans: 2.0625]

2Q. Find the real root of the eqn $x^3 - 9x + 1 = 0$. correct to 3 decimal places. [ans: 2.944]

→ 1) $f(x) = x^3 - 2x - 5 = 0 \Rightarrow f(0) = -5$

$f(1) = -6$

$f(2) = -1$

$f(3) = 16$

$[a,b] = [2,3] \Rightarrow x_1 = 2.5 \Rightarrow f(x_1) = 5.625$

$[a,b] = [2, 2.5] \Rightarrow x_2 = 2.25 \Rightarrow f(x_2) = 1.89$

$[a,b] = [2, 2.25] \Rightarrow x_3 = 2.125 \Rightarrow f(x_3) = 0.345$

$[a,b] = [2, 2.125] \Rightarrow x_4 = 2.0625 \Rightarrow f(x_4) = -0.351$

2) $f(x) = x^3 - 9x + 1 = 0 \Rightarrow f(0) = 1 \quad f(2) = -9$

$f(1) = -7 \quad f(3) = 1$

$[a,b] = [2,3] \Rightarrow x_1 = 2.5 \Rightarrow f(x_1) = -5.875$

$[a,b] = [2.5, 3] \Rightarrow x_2 = 2.75 \Rightarrow f(x_2) = -2.953$

$[a,b] = [2.75, 3] \Rightarrow x_3 = 2.875 \Rightarrow f(x_3) = -1.111$

$[a,b] = [2.875, 3] \Rightarrow x_4 = 2.9375 \Rightarrow f(x_4) = -0.09$

$[a,b] = [2.9375, 3] \Rightarrow x_5 = 2.9687 \Rightarrow f(x_5) = 0.446$

$[a,b] = [2.9375, 2.9687] \Rightarrow x_6 = 2.9531 \Rightarrow f(x_6) = 0.175$

$[a,b] = [2.9375, 2.9531] \Rightarrow x_7 = 2.9453 \Rightarrow f(x_7) = 0.042$

$[2.9375, 2.9453]$

$\Rightarrow x_8 = 2.9414$

$\Rightarrow f(x_8) = -0.02$

$[2.9414, 2.9453]$

$\Rightarrow x_9 = 2.943$

$\Rightarrow f(x_9) = 9.0 \times 10^{-3}$

$[2.9414, 2.943]$

- Newton - Raphson Method :

eg: $f(x) = 0$

(i) a, b such that $f(a)\,f(b) < 0$

(ii) let $x_0 =$ ____ (close to 0)

(iii) $$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Q. Find a real root of the equation $3x = \cos x + 1$ correct to 4 decimal places.

→ let $f(x) = \cos x - 3x + 1$

$\Rightarrow f(0) = 2$

$f(1) = -1.45$

$f(x) = \cos x - 3x + 1$

$f'(x) = -\sin x - 3$

$x_0 = 0$

$f(x_0) = 2$

$f'(x_0) = -3$

$x_1 = 0.666$ $\quad f(x_1) = -0.211 \quad f'(x_1) = -3.61$

$x_2 = 0.6075$ $\quad f(x_2) = -1.42\times10^{-3}$ $\quad f'(x_2) = -3.57$

$x_3 = 0.6071$ $\quad f(x_3) = 5.88\times10^{-6}$ $\quad f'(x_3) = -3.57$

$x_4 = 0.6071$

HW.

1Q. $f(x) = xe^x - 1$ correct upto 4 decimal places. [ans: 0.5671]

2Q. solve $x\log_{10}x = 1.2$ upto 4 decimal places.

→ 1) $f(x) = xe^x - 1$ $\Rightarrow f(0) = -1$

$f(1) = 1.718$

$f'(x) = e^x + xe^x$

$= e^x(1+x)$

$x_0 = 0$

$f(x_0) = -1$ $\quad f'(x_0) = 1$

$x_1 = 1$ $\quad f(x_1) = 1.718$ $\quad f'(x_1) = 5.436$

$x_2 = 0.683$ $\quad f(x_2) = 0.352$ $\quad f'(x_2) = 3.332$

$x_3 = 0.5773$ $\quad f(x_3) = 0.028$ $\quad f'(x_3) = 2.809$

$x_4 = 0.5673$ $\quad f(x_4) = 4.33\times10^{-4}$ $\quad f'(x_4) = 2.763$

$x_5 = 0.5671$ $\quad f(x_5) = -1.196\times10^{-4}$ $\quad f'(x_5) = 2.763$

$x_6 = 0.5671$

2) $f(x) = x\log_{10}x = 1.2$

$f(x) = x\log_{10}x - 1.2$

$f'(x) = \log_{10}x + x\cdot\frac{1}{x\ln 10}$

$f(0) = -1.2$

$f(1) = -1.2$

$f(2) = -0.597$

$x_0 = 2$ $\quad f(x_0) = -0.597$ $\quad f'(x_0) = 0.735$

$x_1 = 2.8122$ $\quad f(x_1) = 0.062$ $\quad f'(x_1) = 0.883$

$x_2 = 2.7419$ $\quad f(x_2) = 1.093 \times 10^{-3}$ $\quad f'(x_2) = 0.872$

$x_3 = 2.7406$ $\quad f(x_3) = -4.02 \times 10^{-5}$ $\quad f'(x_3) = 0.872$

$x_4 = 2.7406$

Q. Find a positive value of $4\sqrt{5}$ using Newton-Raphson method correct to 2 decimal places.

→ let $x = 4\sqrt{5}$ $\Rightarrow x^4 = 5$; $x^4 - 5 = 0$

$f(x) = x^4 - 5$ $\quad f'(x) = 4x^3$

$f(0) = -5$

$f(1) = -4$ $\quad x_0 = 1.5 \Rightarrow f(x_0) = 0.062$ $\quad f'(x_0) = 6.912$

$f(2) = 11$ $\quad x_1 = 1.495 \Rightarrow f(x_1) = -4.863 \times 10^{-3}$ $\quad f'(x_1) = 13.365$

$x_2 = 1.495$

•) Secant Method

$$x_{n+1} = x_n - \frac{f(x_n)\,(x_n - x_{n-1})}{f(x_n) - f(x_{n-1})}$$

OR

$$x_{n+1} = \frac{x_{n-1}\, f(x_n) - x_n\, f(x_{n-1})}{f(x_n - f(x_{n-1})}$$

Q. Find the root of $x^3 - 5x + 1 = 0$ in interval $(0,1)$

here,

$x_0 = 0$

$x_1 = 1$

$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)}$$

$f(x_0) = 1$

$f(x_1) = -3$

$$= \frac{0[-3] - -1[1]}{-3-1} = \frac{-1}{-4} = 0.25$$

$f(x_2) = -0.23$

$$x_3 = \frac{1(-0.23) - 0.25(-3)}{-0.23 - (-3)} = 0.187 \qquad f(x_3) = 0.071$$

$$x_4 = \frac{0.25(0.071) - 0.187(-0.23)}{0.071 - (-0.23)} = 0.201 \qquad f(x_4) \doteq 3.1 \times 10^{-3}$$

$$x_5 = \frac{x_3 f(x_4) - x_4 f(x_3)}{f(x_4) - f(x_3)} = \frac{0.187 \times 3.1\times10^{-3} - 0.201 \times 0.071}{3.1\times10^{-3} - 0.071}$$

$= 0.201$ $\qquad f(x_5) = 2.4 \times 10^{-6} \approx 0$

$\therefore$ root $= 0.201$

HW

1Q. Find the real root of eqn $x^3 + x^2 - 1$. perform 4 iteration using secant method [ans: 0.7533]

2Q. Find the root of $e^x - \sin x = 0$ using secant method. correct to 3 decimal places.

→ 1) $f(x) = x^3 + x^2 - 1$

$f(0) = -1 \Rightarrow x_0 = 0$

$f(1) = 1 \Rightarrow x_1 = 1$

$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)} = \frac{0 \times 1 - (1 \times -1)}{1 - (-1)} = \frac{1}{2} = 0.5$$

$f(x_2) = -0.625$; $x_3 = \frac{x_1 f(x_2) - x_2 f(x_1)}{f(x_2) - f(x_1)} = \frac{1 \times -0.625 - (0.5 \times 1)}{-0.625 - 1}$

$x_3 = 0.6923 \qquad f(x_3) = -0.1889$

$$x_4 = \frac{x_2 f(x_3) - x_3 f(x_2)}{f(x_3) - f(x_2)} = \frac{0.5 \times -0.1889 - (0.6923 \times -0.625)}{-0.1889 - (-0.625)}$$

$x_4 = 0.7755 \qquad f(x_4) = 0.0677$

$$x_5 = \frac{x_3 f(x_4) - x_4 f(x_3)}{f(x_4) - f(x_3)} = \frac{0.6923 \times 0.0677 - (0.7755 \times -0.1889)}{0.0677 - (-0.1889)} = 0.7535$$

$f(x_5) = 0.0044 \approx 0 \qquad \therefore$ root $= 0.7535$

2) $f(x) = e^x - \sin x$

$f(0) = 1 \qquad f(-3) = 0.190$

$f(1) = 1.876 \qquad f(-4) = -0.738$

$\therefore x_0 = -4$

$x_1 = -3$

$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)} = \frac{-4 \times 0.190 - (-3 \times -0.738)}{0.190 - (-0.738)} = -3.204$$

$f(x_2) = -0.021 \quad ; \quad x_3 = \frac{x_1 f(x_2) - x_2 f(x_1)}{f(x_2) - f(x_1)} = \frac{-3 \times -0.021 - (-3.204 \times .19)}{-0.021 - 0.190}$

$x_3 = -3.183 \qquad f(x_3) = 0.0000655 \approx 0$

$x_4 = -3.183$

$\therefore$ root $= -3.183$

• Method of Successive iteration [Fixed Point Iteration Method]

Find $x = \phi(x)$ s. $|\phi'(x)| < 1$

$x_{n+1} = \phi(x_n)$

$x_1 = \phi(x_0)$

Q. Find root of $x^3 + x^2 - 1 = 0$ correct upto 5 decimal places.

→ $f(x) = x^3 + x^2 - 1$

$f(0) = -1 \qquad x_0 = 0.5 \qquad [a, b] = [0, 1]$

$f(1) = 1$

To find $\phi(x)$

$x^3+x^2-1=0 \Rightarrow x^3+x^2=1 \Rightarrow x^2(x+1)=1 \Rightarrow x^2=\frac{1}{x+1}$

$x=\frac{1}{\sqrt{x+1}}$

$x_1=\phi(x_0)=\frac{1}{\sqrt{0.5+1}}=0.81650$

$x_2=\phi(x_1)=\frac{1}{\sqrt{0.81650+1}}=0.74196$

$x_3=\phi(x_2)=\frac{1}{\sqrt{0.74196+1}}=0.75767$

$x_4=\phi(x_3)=\frac{1}{\sqrt{1.75767}}=0.75428$

$x_5=\phi(x_4)=\frac{1}{\sqrt{1.75428}}=0.75501$

$x_6=\phi(x_5)=\frac{1}{\sqrt{1.75501}}=0.75485$

$x_7=\phi(x_6)=\frac{1}{\sqrt{1.75485}}=0.75488$

$x_8=\phi(x_7)=\frac{1}{\sqrt{1.75488}}=0.75488$

Q. Find the root of $f(x)=2x^3-2x-5$ correct upto 3 decimal places

→ $f(x)=2x^3-2x-5$ $\quad f(0)=-5$ $\quad f(2)=7$ $\quad [a,b]=[1,2]$

$f(1)=-5$

$x_0=1.5$

$\phi(x)$

$2x^3-2x-5 \Rightarrow 2x^3-2x=5 \Rightarrow 2x(x^2-1)=5 \Rightarrow 2x=\frac{5}{x^2-1}$

$x=\frac{5}{2(x^2-1)}$

$2x^3-2x-5 \Rightarrow 2x\,x^2-1=\frac{5}{2x} \Rightarrow x=\sqrt{\frac{5}{2x}+1}$

$x_1 = \phi(x_0) = \sqrt{\frac{5}{2x1.5} + 1} = 1.633$

$x_2 = \phi(x_1) = 1.591$ $\quad x_3 = \phi(x_2) = 1.604$

$x_4 = \phi(x_3) = \underline{1.600}$ $\quad x_5 = 1.601$ $\quad x_6 = 1.600$

System of Linear Equations

Gauss-Seidel Iteration Method

$a_1x + b_1y + c_1z = d_1$

$a_2x + b_2y + c_2z = d_2$

$a_3x + b_3y + c_3z = d_3$

$x = \frac{1}{a_1}[d_1 - b_1y - c_1z]$; $|a_1| \geq |b_1| + |c_1|$

$y = \frac{1}{b_2}[d_2 - a_2x - c_2z]$; $|b_2| \geq |a_2| + |c_2|$

$z = \frac{1}{c_3}[d_3 - a_3x - b_3y]$; $|c_3| \geq |a_3| + |b_3|$

Q. Solve the following system of eqn using Gauss-Seidel Iteration method.

$10x + 2y + z = 9$ —① Perform 3 Iteration.

$x + 10y - z = -22$ —②

$-2x + 3y + 10z = 22$ —③

→ $x = \frac{1}{10}[9 - 2y - z]$; $y = \frac{1}{10}[-22 - x + z]$; $z = \frac{1}{10}[22 + 2x - 3y]$

assume $x^{(0)} = y^{(0)} = z^{(0)} = 0$

Iteration 1

$x_1 = \frac{1}{10}[9 - 2y_0 - z_0] = \frac{9}{10} = 0.9$

$y_1 = \frac{1}{10}[-22 - x_1 + z_0] = \frac{1}{10}[-22 - 0.9 + 0]$

$= -2.29$

$z_1 = \frac{1}{10}[22 + 2x_1 - 3y_1] = \frac{1}{10}[22 + 2 \times 0.9 - 3 \times (-2.29)] = 3.067$

Iteration 2

$$x_2 = \frac{1}{10}[9 - 2y_1 - z_1] = \frac{1}{10}[9 - 2\times(-2.29) - 3.067] = 1.051$$

$$y_2 = \frac{1}{10}[-22 - x_2 + z_1] = \frac{1}{10}[-22 - 1.051 + 3.067] = -1.99$$

$$z_2 = \frac{1}{10}[22 + 2x_2 - 3y_2] = \frac{1}{10}[22 + (2\times 1.051) - (3\times -1.99)] = 3.007$$

Iteration 3

$$x_3 = \frac{1}{10}[9 - 2y_2 - z_2] = \frac{1}{10}[9 - (2\times -1.99) - (3.007)] = 0.997$$

$$y_3 = \frac{1}{10}[-22 - 0.997 + 3.007] = -1.999$$

$$z_3 = \frac{1}{10}[22 + (2\times 0.997) - (3\times -1.999)] = 2.999$$

$x = 1$, $y = -2$, $z = 3$

Q. Solve the system of equation

$$4x + 11y - z = 33$$
$$6x + 3y + 12z = 36$$
$$8x - 3y + 2z = 20$$

→

$$8x - 3y + 2z = 20 \quad \text{——} (1)$$
$$4x + 11y - z = 33 \quad \text{——} (2)$$
$$6x + 3y + 12z = 36 \quad \text{——} (3)$$

$$x = \frac{1}{8}[20 + 3y - 2z]$$

$$y = \frac{1}{11}[33 - 4x + z]$$

$$z = \frac{1}{12}[36 - 6x - 3y]$$

assume $x_0 = y_0 = z_0 = 0$

Iteration 1

$$x_1 = \frac{1}{8}[20 + 3y_0 - 2z_0] = 2.5$$

$$y_1 = \frac{1}{11}[33 - 4x_1 + z_0] = \frac{1}{11}[33 - 4\times 2.5 + 0] = 2.09$$

$$z_1 = \frac{1}{12}[36 - 6x_1 - 3y_1] = \frac{1}{12}[36 - (6\times 2.5) - (3\times 2.09)] = 1.227$$

Iteration 2

$$x_2 = \frac{1}{8}[20 + (3\times 2.09) - (2\times 1.227)] = 2.977$$

$y_2 = \frac{1}{11}[33 - (4 \times 2.977) + 1.227] = 2.029$

$z_2 = \frac{1}{12}[36 - (6 \times 2.977) - (3 \times 2.029)] = 1.004$

Iteration 3

$x_3 = \frac{1}{8}[20 + (3 \times 2.029) - (2 \times 1.004)] = 3.01$

$y_3 = \frac{1}{11}[33 - (4 \times 3.01) + 1.004] = 1.997$

$z_3 = \frac{1}{12}[36 - (6 \times 3.01) - (3 \times 1.997)] = 0.996$

Iteration 4

$x_4 = \frac{1}{8}[20 + (3 \times 1.997) - (2 \times 0.996)] = 3$

$y_4 = \frac{1}{11}[33 - (4 \times 3) + 0.996] = 2$

$z_4 = \frac{1}{12}[36 - (6 \times 3) - (3 \times 2)] = 1$

$\therefore x = 3 \;,\; y = 2 \;,\; z = 1$

HW

Q. (I)
$6x - y - z = 19$
$3x + 4y + z = 26$
$x + 2y + 6z = 22$

(II)
$5x + 2y + z = 12$
$x + 4y + 2z = 15$
$x + 2y + 5z = 20$

Interpolation

The process of finding intermediate values of a function from a given set of values. i.e finding $f(x)$ for any value of x in the interval $[x_0, x_n]$.

If the values of x are at equal intervals we use Newton's forward or backward formula. Lagranges formula can be used in both cases.

Lagrange's Interpolation

Let $x : x_0, x_1, x_2, \cdots x_n$

$f(x) = y : y_0, y_1, y_2, \cdots y_n$

To find $f(x)$, $x \in (x_0\, x_n)$

$$f(x) = \frac{(x-x_1)(x-x_2)(x-x_3)\cdots(x-x_n)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)\cdots(x_0-x_n)} \times y_0$$

$$+ \frac{(x-x_0)(x-x_2)(x-x_3)\cdots(x-x_n)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)\cdots(x_1-x_n)} \times y_1$$

$$+ \frac{(x-x_0)(x-x_1)(x-x_3)\cdots(x-x_n)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)\cdots(x_2-x_n)} \times y_2$$

Q.

	x_0	x_1	x_2	x_3
x	5	6	9	11
y	12	13	14	16
	y_0	y_1	y_2	y_3

Find for $x = 10$ or Find $y(10)$?

→

$$f(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} \times y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} \times y_1 +$$

$$\frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} \times y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} \times y_3$$

$$f(10) = \frac{(10-6)(10-9)(10-11)}{(5-6)(5-9)(5-11)} \times 12 + \frac{(10-5)(10-9)(10-11)}{(6-5)(6-9)(6-11)} \times 13$$

$$+ \frac{(10-5)(10-6)(10-11)}{(9-5)(9-6)(9-11)} \times 14 + \frac{(10-5)(10-6)(10-9)}{(11-5)(11-6)(11-9)} \times 16$$

$$= 0.167 \times 12 + (-0.333 \times 13) + 0.833 \times 14 + 0.333 \times 16$$

$= \underline{14.665}$

Q. Find $f(x)$ from the table

x	0	1	2	5
y	2	3	12	147

& Find $f(3)$

$$\rightarrow f(x) = \frac{(x-1)(x-2)(x-5)}{(0-1)(0-2)(0-5)} \times 2 + \frac{(x-0)(x-2)(x-5)}{(1-0)(1-2)(1-5)} \times 3 + \frac{(x-0)(x-1)(x-5)}{(2-0)(2-1)(2-5)} \times 12$$

$$+ \frac{(x-0)(x-1)(x-2)}{(5-0)(5-1)(5-2)} \times 147$$

$$= \frac{(x-1)(x-2)(x-5)}{-10} \times 2 + \frac{(x-0)(x-2)(x-5)}{4} \times 3 + \frac{(x-0)(x-1)(x-5)}{-6} \times 12$$

$$+ \frac{(x-0)(x-1)(x-2)}{60} \times 147$$

$$\Rightarrow \frac{(x^2-3x+2)(x-5)}{-10} \times 2 + \frac{(x^2-2x)(x-5)}{4} \times 3 + \frac{(x^2-x)(x-5)}{-6} \times 12$$

$$+ \frac{(x^2-x)(x-2)}{60} \times 147$$

$$\Rightarrow \frac{x^3-3x^2+2x-5x^2+15x-10}{-5} + \frac{3(x^3-2x^2-5x^2+10x)}{4} -$$

$$2(x^3-x^2-5x^2+5x) + \frac{49(x^3-x^2-2x^2+2x)}{20}$$

$$\Rightarrow \frac{x^3-8x^2+17x-10}{-5} + \frac{3x^3-21x^2+30x}{4} - 2x^3+12x^2-10x + \frac{49x^3-147x^2+98x}{20}$$

$$\Rightarrow x^3\left(\frac{-1}{5}+\frac{3}{4}-2+\frac{49}{20}\right) + x^2\left(\frac{8}{5}-\frac{21}{4}+12-\frac{147}{20}\right) + x\left(\frac{-17}{5}+\frac{30}{4}-10+\frac{98}{20}\right)+2$$

$$\Rightarrow x^3(1) + x^2(1) + x(-1) + 2$$

$$f(x) = x^3+x^2-x+2 \qquad f(3) = 3^3+3^2-3+2 = \underline{35}$$

nw
1Q. Find the value of $\tan 33^\circ$ by using Lagrange's formula for Interpolation from the following table.

x	30°	32°	35°	36°
$\tan x$	0.5774	0.6249	0.7002	0.7813

[0.6494]

2Q. A second degree polynomial passes through the points $(1,-1)$ $(2,-1)$ $(3,1)$ $(4,5)$. Find the polynomial $f(x)$. Also find $f(1.2)$. $[f(x) = x^2 - 3x + 1]$

Newton's Forward Interpolation

Forward difference table (Δ):

x	y	Δy	$\Delta^2 y$
x_0	y_0	$\Delta y_0 = y_1 - y_0$	
x_1	y_1	$\Delta y_1 = \Delta y_2 - \Delta y_1$	
x_2	y_2	$\Delta y_2 = \Delta y_3 - \Delta y_2$	
⋮	⋮		
x_n	y_n	$\Delta y_{n-1} = y_n - y_{n-1}$	

Formula : $$f(x) = y_0 + u\,\Delta y_0 + \frac{u(u-1)}{2!}\Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!}\Delta^3 y_0 + \cdots$$

$$u = \frac{x - x_0}{h}$$

x = finding value
x_0 = initial value
$h = x_1 - x_0 = x_2 - x_1$ (difference of x_i)

Q. Compute $f(1.5)$ from $f(1) = -1$, $f(2) = -1$, $f(3) = 1$ and $f(4) = 5$ by using Newton's forward Interpolation.

→ step1 : construct forward difference table

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
1	-1			
		0		
2	-1		2	
		2		0
3	1		2	
		4		
4	5			

step 2: formula, $u = \frac{x - x_0}{h} = \frac{1.5 - 1}{1} = 0.5$

$\Rightarrow f(x) = y_0 + u\Delta y_0 + \frac{u(u-1)}{2!}\Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!}\Delta^3 y_0 + \cdots$

$f(1.5) = -1 + (0.5 \times 0) + \frac{0.5(0.5-1)}{2!} \times 2 + \frac{0.5(0.5-1)(0.5-2)}{3!} \times 0$

$= \underline{\underline{-1.125}}$

Q. Find $f(0.5)$ using the values $f(0) = 1$, $f(1) = 2$, $f(2) = 1$ & $f(3) = 10$. Also find $\Delta^3 f(x)$

→

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
0	1			
		1		
1	2		−2	
		−1		12
2	1		10	
		9		
3	10			

$u = \frac{0.5 - 0}{1} = 0.5$

$f(0.5) = 1 + (0.5 \times 1) + \frac{0.5(0.5-1)}{2!} \times -2 + \frac{0.5(0.5-1)(0.5-2)}{3!} \times 12$

$= 1.5 + 0.25 + 0.750$

$= \underline{\underline{2.5}}$

$\therefore \Delta^3 f(x) = \underline{\underline{12}}$

Newton's Backward Interpolation

Backward difference table (∇):

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
x_0	y_0				
		∇y_1			
x_1	y_1		$\nabla^2 y_2$		
		∇y_2		$\nabla^3 y_3$	
x_2	y_2		$\nabla^2 y_3$		$\nabla^4 y$
		∇y_3		$\nabla^3 y_4$	
x_3	y_3		$\nabla^2 y_4$		
		∇y_4			
x_4	y_4				

Formula: $f(x) = y_n + u\nabla y_n + \frac{u(u+1)}{2!}\nabla^2 y_n + \frac{u(u+1)(u+2)}{3!}\nabla^3 y_n + \cdots$

$u = \frac{x - x_n}{h}$

x = finding value
x_n = last value of x
h = diff of x values

Q. The population of a town is given. Estimate the population of the year 1925.

Year :	1891	1901	1911	1921	1931
Population (In 1000's) :	46	66	81	93	101

→

steps	x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
	1891	46				
			20			
	1901	66		-5		
			15		2	
	1911	81		-3		-3
			12			
	1921	93		-4	-1	
			8			
	1931	101				

$$u = \frac{x - x_n}{h} = \frac{1925 - 1931}{10} = -0.6$$

$$f(x) = y_n + u\nabla y_n + \frac{u(u+1)}{2!}\nabla^2 y_n + \frac{u(u+1)(u+2)}{3!}\nabla^3 y_n + \dots$$

$$f(1925) = 101 + (-0.6 \times 8) + \frac{-0.6(-0.6+1)}{2!} \times -4 + \frac{-0.6(-0.6+1)(-0.6+2)}{3!} \times -1$$

$$+ \frac{-0.6(-0.6+1)(-0.6+2)(-0.6+3)}{4!} \times -3$$

$$= 101 + -4.8 + 0.48 + 0.056 + 0.101 = 96.837 \approx 97$$

Q. Find the first and last term.

x :	3	4	5	6	7	8	9
y :	4.8	8.4	14.5	23.6	36.2	52.8	73.9

→ To find $f(1)$ ⇒ Forward interpolation and $f(10)$ ⇒ backward Interpolation

x	y	Δy or ∇y	$\Delta^2 y$ on $\nabla^2 y$	$\Delta^3 y$ or $\nabla^3 y$	$\Delta^4 y$ or $\nabla^4 y$
3	4.8				
		3.6			
4	8.4		2.5		
				0.5	

5	14·5	6·1		0·5	
			3		0
		9·1		0·5	
6	23·6		3·5		0
		12·6		0·5	
7	36·2		4		0
		16·6		0·5	
8	52·8		4·5		
		21·1			
9	73·9				

$u = \frac{1-3}{1} = \frac{-2}{1} = -2$

$$f(1) = 4\cdot8 + (-2 \times 3\cdot6) + \frac{-2(-2-1)}{2!} \times 2\cdot5 + \frac{-2(-2-1)(-2-2)}{3!} \times 0\cdot5$$

$$= 4\cdot8 + (-7\cdot2) + 7\cdot5 + (-2) = \underline{\underline{3\cdot1}}$$

$u = \frac{10-9}{1} = 1$

$$f(10) = 73\cdot9 + 1 \times 21\cdot1 + \frac{1(1+1)}{2!} \times 4\cdot5 + \frac{1(1+1)(1+2)}{3!} \times 0\cdot5$$

$$= 73\cdot9 + 21\cdot1 + 4\cdot5 + 0\cdot5 = \underline{\underline{100}}$$

Newton's Divided Difference Interpolation

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
x_0	y_0	$\Delta y_0 = \frac{y_1 - y_0}{x_1 - x_0}$	$\Delta^2 y_0 = \frac{\Delta y_1 - \Delta y_0}{x_2 - x_0}$	$\Delta^3 y_0 = \frac{\Delta^2 y_1 - \Delta^2 y_0}{x_3 - x_0}$
x_1	y_1		$\Delta^2 y_1 = \frac{\Delta y_2 - \Delta y_1}{x_3 - x_1}$	
x_2	y_2			
x_3	y_3			
x_4	y_4			

Formula: $f(x) = y_0 + (x - x_0)\Delta y_0 + (x - x_0)(x - x_1)\Delta^2 y_0 + (x - x_0)(x - x_1)(x - x_2)\Delta^3 y_0$

$+ \cdots\cdots$

Q. Given the values. Evaluate $f(9)$ using Newton's divided difference formula.

x :	5	7	11	13	17
$f(x)$:	150	392	1452	2366	5202

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
5	150	121		
7	392	265	24	1
11	1452	457	32	1
13	2366	709	42	
17	5202			

$$f(9) = 150 + (9-5) \times 121 + (9-5)(9-7) \times 24 + (9-5)(9-7)(9-11) \times 1$$

$$= 150 + 484 + 192 + (-16)$$

$$= \underline{\underline{810}}$$

Q. Find the fn $f(x)$ from following table using Newton's divided difference formula

x	1	2	7	8
$f(x)$	1	5	5	4

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
1	1			
2	5	4	-0.667	0.071
7	5	0	-0.167	
8	4	-1		

$$f(x) = 1 + (x-1) \times 4 + (x-1)(x-2) \times -0.667 + (x-1)(x-2)(x-7) \times 0.071$$

$$= 1 + 4x - 4 + (x^2 - 2x - x + 2) \times -0.667 + (x^3 - 2x^2 - x^2 + 2x - 7x^2 + 14x - 7x - 14) \, 0.071$$

$$= 1 + 4x - 4 + (x^2 - 3x + 2) - 0.667 + (x^3 - 3x^2 - 5x - 21x) \, 0.071$$

$$= 1 + 4x - 4 - 0.667x^2 + 2.001x - 1.334 + 0.071x^3 - 0.213x^2 - 0.355x - 1.491x$$

$$= 1 + 4 - 1.334 + 4x + 2.001x - 0.355x - 1.491x - 0.667x^2 - 0.213x^2 + 0.071x^3$$

$$= 3.666 + 4.155x^2 - 0.880x^2 + 0.071x^3$$

$$= 0.071x^3 - 0.88x^2 + 4.155x + 3.66$$

$(0.07x^2 - 1.36x^2 + 7.611x + 5.31)$

Least square method

When data points donot perfectly align, they create residuals (errors). The least square method targets the line where the sum of these squared errors is at its absolute minimum.

[curve fitting]

(i) Straight line curve fitting (using least square method)

$$y = a + bx \quad \text{——— ①}$$

The normal eqns are : $\Sigma y = Na + b\Sigma x$ ——— ② (sum of ①)

$$\Sigma xy = a\Sigma x + b\Sigma x^2 \quad \text{——— ③ (sum of ② } \times x)$$

By solving ② and ③, find values of a and b.

Q. Using method of least square, find best fitting straight line to the following data.

x	1	2	3	4	5
y	1	3	5	6	5

→

x	y	xy	x^2
1	1	1	1
2	3	6	4
3	5	15	9
4	6	24	16
5	5	25	25
Total: 15	20	71	55

$\Sigma x = 15$, $\Sigma y = 20$, $\Sigma xy = 71$,

$\Sigma x^2 = 55$

$N = 5$

$20 = 5a + 15b$ ——— ②

$71 = 15a + 55b$ ——— ③

$a = 0.7$ $\quad b = 1.1$

$\therefore y = a + bx \Rightarrow y = 0.7x + 1.1x$

Q. The temperature 'T' and length 'L' of a heated road are given below. If $L = a_0 + a_1 T$. find the best values for a_0 and a_1.

T	20	30	40	50	60	70
L	800.3	800.4	800.6	800.7	800.9	801

→ Let x represent T and y represent L

$y = a + bx$ — ① $\quad \Sigma y = Na + b\Sigma x$ — ② $\quad \Sigma xy = a\Sigma x + b\Sigma x^2$ — ③

T	L	T^2	TL
20	800·3	400	16006
30	800·4	900	24012
40	800·6	1600	32024
50	800·7	2500	40035
60	800·9	3600	48054
70	801	4900	56070
270	4803·9	13900	216201

Total Σ

N=6

Let,

$x = T$

$y = L$

resulting ①, ② and ③

$L = a_0 + a_1 T$ — ①

$L = 800 + 0.0146T$

② $\Rightarrow \Sigma L = Na_0 + a_1 \Sigma T$

③ $\Rightarrow \Sigma TL = a_0 \Sigma T + a_1 \Sigma T^2$

substituting,

② $\Rightarrow 4803.9 = 6a_0 + 270a_1$

③ $\Rightarrow 216201 = 270a_0 + 13900a_1$

$a_0 = 799.99 \approx 800$

$a_1 = 0.01457 \approx 0.0146$

substituting in ①

① $\Rightarrow L = 800 + 0.0146T$

(ii) Second degree Parabola curve fitting

$y = a + bx + cx^2$ — ①

The normal equations are: $\Sigma y = Na + b\Sigma x + c\Sigma x^2$ — ②

$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$ — ③

$\Sigma x^2 y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4$ — ④

Solving eqns ②, ③ and ④, find the values a, b and c and substitute in ①.

Q. Fit a second degree polynomial to the following data by the method of least square.

x	1	2	3	4	5	6	7
y	-5	-2	5	16	31	50	73

→

x	y	x^2	x^3	x^4	xy	x^2y
1	-5	1	1	1	-5	-5
2	-2	4	8	16	-4	-8
3	5	9	27	81	15	45
4	16	16	64	256	64	256
5	31	25	125	625	155	775
6	50	36	216	1296	300	1800
7	73	49	343	2401	511	3577
Σ total	28	168	140	784	4676	1036	6440

N=7

$y = a + bx + cx^2$ — ①

$\Sigma y = Na + b\Sigma x + c\Sigma x^2$ — ②

$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$ — ③

$\Sigma x^2 y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4$ — ④

② → $168 = 7a + 28b + 140c$

③ → $1036 = 28a + 140b + 784c$

④ → $6440 = 140a + 784b + 4676c$

$a = -4, \quad b = -3, \quad c = 2$

◉ substituting in ① ⇒ $y = -4 - 3x + 2x^2$

⇒ $y = 2x^2 - 3x - 4$

Q. Fit a second degree curve to the following data and find the value of y when $x = 7$.

x	1	2	3	4	5
y	7	25	53	91	139

$N = 5$

	x	y	x^2	x^3	x^4	xy	x^2y
	1	7	1	1	1	7	7
	2	25	4	8	16	50	100
	3	53	9	27	81	159	477
	4	91	16	64	256	364	1456
	5	139	25	125	625	695	3475
Σ:	15	315	55	225	979	1275	5515

$y = a + bx + cx^2$ —— ①

$\Sigma y = Na + b\Sigma x + c\Sigma x^2$ —— ②

$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$ —— ③

$\Sigma x^2y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4$ —— ④

② ⇒ $315 = 5a + 15b + 55c$

③ ⇒ $1275 = 15a + 55b + 225c$

④ ⇒ $5515 = 55a + 225b + 979c$

solving ②, ③ and ④,

$a = -1$, $b = 3$, $c = 5$

① ⇒ $y = -1 + 3x + 5x^2$ ⇒ $y = 5x^2 + 3x - 1$

when $x = 7$ ⇒ $y = 5 \times (7)^2 + 3 \times 7 - 1 = 265$

HW

Q. (1)

x	-2	-1	0	1	2
y	19	27	29	33	29

(2)

x	0	1	2	3	4
y	1	1·8	1·3	2·5	6·3

(iii) Exponential curve fitting

$y = ae^{bx}$ —— (I)

applying log on both sides ⇒ $\log y = \log a + \log e^{bx}$

⇒ $\log y = \log a + bx \log e$

note: $\log_e e = 1$

⇒ $\log y = \log a + bx$ —— ①

comparing ① with eqn of straight line i.e $Y = A + bx$, where $Y = \log y$, $A = \log a$

$Y = A + bx$ —— ②

the normal eqns are: $\Sigma Y = NA + b\Sigma x$ —— ③ , $\Sigma xY = A\Sigma x + b\Sigma x^2$ —— ④

solving ③ and ④ ⇒ find the values of A & b and substitute in (I).

Since $A = \log a \Rightarrow a = \text{Antilog}(A)$

Q. Fit the curve $y = ae^{bx}$ to the following data.

x	0	2	4
y	5.012	10	31.62

x	y	$Y = \log_e y$ (ln)	x^2	xY
0	5.012	1.611	0	0
2	10	2.3	4	4.6
4	31.62	3.45	16	13.8
6		7.36	20	18.4

$y = ae^{bx}$ —— (I)

$Y = A + bx$ where, $Y = \log y$, $A = \log a$

$\Sigma Y = NA + b\Sigma x$ —— (2)

$\Sigma xy = A\Sigma x + b\Sigma x^2$ —— (3)

$N = 3$

$7.36 = 3A + 6b$ —— (2)

$18.4 = 6A + 20b$ —— (3)

$A = 1.533$ $\quad b = \underline{0.46}$

$a = \text{antilog}(A) = e^{1.533} = \underline{4.63}$

(I) ⇒ $y = 4.63\, e^{0.46x}$

$V = qe^{bt}$

Q. Fit the curve of the form $y = ae^{bx}$

x	77	100	185	239	285
y	2.4	3.4	7.0	11.1	19.6

x	y	$Y = \log_e y$	x^2	xY
77	2.4	0.87	5929	66.99
100	3.4	1.22	10000	122
185	7.0	1.94	34225	358.9
239	11.1	2.4	57121	573.6
285	19.6	2.97	81225	846.45
886		9.4	188500	1967.94

$y = ae^{bx}$ —— (I)

$N = 5$

$\Sigma Y = NA + b\Sigma x$ —— (2)

$\Sigma xy = A\Sigma x + b\Sigma x^2$ —— (3)

$9.4 = 5A + 886b$ —— (2)

$1967.94 = 886A + 188500b$ —— (3)

$A = 0.179$ $\quad b = \underline{0.0095}$

$a = e^{0.179} = \underline{1.196}$

(I) ⇒ $y = 1.196\, e^{0.0095x}$